Jeffrey Robinson

Senior Full Stack Engineer

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SUMMARY

Experienced Senior Full Stack Engineer with over 10 years of expertise in designing and developing scalable, high-performance applications across various domains. Proficient in **Python**, **Java**, **Golang**, **React**, and **Node**.**js**, with strong hands-on experience in cloud-native architectures, microservices, and CI/CD pipelines. Adept at optimizing system performance through database tuning, containerization, and infrastructure automation. Proven success in leading cross-functional teams, implementing secure authentication systems, and driving continuous improvement in development and deployment processes. Extensive background in AWS, PostgreSQL, MongoDB, and container orchestration with Docker and Kubernetes, delivering reliable and cost-effective solutions in fast-paced environments.

EXPERIENCE

Oct 2021 – Jan 2025 **Senior Full Stack Engineer,** Accenture **|** location: Orlando, FL

- Led performance optimization initiatives for **Python** applications, focusing on identifying and eliminating bottlenecks in database queries and memory usage. This resulted in significant improvements in overall application speed and performance.
- Designed and built high-performance, low-latency **APIs** using **Golang** in a distributed systems environment, leading to improved request throughput and overall system responsiveness.
- Spearheaded the design and development of **RESTful APIs** using **Flask**, enabling seamless communication between various microservices, which improved data flow and overall application architecture.
- Built and optimized real-time **APIs** to serve machine learning model predictions using **Flask** and **FastAPI**, enhancing real-time decision-making capabilities in production environments.
- Designed and deployed **RESTful APIs** with **Python** and **Azure API Management** to provide secure, scalable interactions between microservices and external clients, ensuring high availability and efficiency.
- Developed and maintained **Python** applications using **SQLAlchemy** and **SQLModel** to interact with **PostgreSQL** databases, ensuring efficient query execution and data retrieval.
- Designed and optimized database schemas using **SQLAlchemy ORM**, streamlining data relationships and minimizing complex joins to improve query performance in **Postgres**.
- Led the migration of legacy **Django** backend services to **FastAPI**, improving application performance by reducing request latency and enhancing overall response times, leveraging FastAPI's asynchronous capabilities.
- Refactored and optimized existing **Django** views and database queries into **FastAPI** endpoints, ensuring minimal disruption to the user experience while maintaining compatibility with the existing data models.
- Implemented modern, asynchronous **Python** features in **FastAPI**, improving the efficiency of I/O-bound operations such as **API** calls and database interactions, resulting in significant throughput increases.

- Conducted thorough performance benchmarking between **Django** and **FastAPI**, identifying key areas where **FastAPI** provided improved speed and scalability, and used this data to drive decision-making for further optimizations.
- Collaborated with cross-functional teams to optimize cloud-based **Python** applications on **AWS** using services like **EC2**, **S3**, and **Lambda**, leading to a reduction in infrastructure costs through more efficient resource management.
- Configured and deployed applications on **Amazon Elastic Kubernetes Service (EKS)**, leveraging **Kubernetes** for container orchestration, ensuring seamless scaling and high availability of services.
- Collaborated with development teams to ensure smooth integration of **Python**-based services into the **Kubernetes** cluster, addressing potential issues related to resource allocation, networking, and logging.
- Developed a comprehensive containerization strategy using **Docker** and **Kubernetes**, allowing seamless deployment and scaling of **Python** microservices across multi-cloud environments, which improved performance and resource utilization.
- Optimized **CI/CD** pipelines for **Python** applications using **GitLab CI**, **Jenkins**, and **GitHub Actions**. This streamlined the release process, improving development cycles while maintaining high-quality standards across environments.
- Designed and optimized complex **SQL** queries and stored procedures to improve data processing workflows, significantly reducing query execution times and improving overall data retrieval efficiency.
- Led the migration from **SQL-based** databases to **MongoDB**, which improved data retrieval speed for large datasets by leveraging MongoDB's flexible schema and powerful indexing features.
- Enhanced **MongoDB** database performance by optimizing replica set and sharding configurations, resulting in increased availability and scalability and minimizing downtime during production incidents.
- Led the creation of an automated **CI/CD pipeline** leveraging **Docker**, **Jenkins**, and **Kubernetes**, improving release cycles by automating builds, testing, and deployment, ensuring high-quality software delivery.
- Implemented **dependency injection** (DI) in a .NET Core application, improving testability and maintainability by decoupling services and allowing for easier unit testing with mocked dependencies.
- Designed and developed a modular **product architecture** using dependency injection in .NET, enabling the addition of new features without major changes to the core codebase, ensuring better scalability and flexibility.
- Integrated **NGINX** with **Python**-based applications (**Flask**/**Django**) to implement reverse proxying, caching, and session management, which reduced response times and offloaded static content handling, improving overall performance.
- Integrated **PyTest** into **CI** pipelines using **Jenkins** and **GitLab CI**, automating the execution of tests, code quality checks, and report generation, which minimized manual testing efforts and accelerated release cycles.
- Integrated complex business logic into the **Entity Framework** (EF) Core layer, ensuring smooth data validation and seamless interaction with SQL Server databases, while leveraging EF Core's migration capabilities for efficient schema changes.
- Optimized data access performance by implementing asynchronous methods and optimizing **LINQ** queries in .NET applications, reducing response times and improving user experience in high-traffic systems.
- Managed end-to-end deployment pipelines using **Ansible**, simplifying the continuous delivery process and reducing deployment times while improving release predictability for **Python** application deployments.
- Optimized **CircleCI** workflows by implementing caching and parallel job execution strategies, reducing build times and speeding up the overall **CI/CD** process for **Python** applications.
- Collaborated with product and engineering teams to define **JIRA** epics, stories, and tasks, ensuring clear project scope and requirements, which enhanced sprint planning and reduced scope creep.

- Worked with cross-functional teams to integrate **Mendix**-based applications with **Python**-driven analytics tools, enabling real-time data analysis and reporting for business stakeholders, improving data-driven decision-making.
- Contributed to the development of a real-time analytics platform for engineering teams using **Plotly.js** and **WebSocket** technology. This allowed dynamic, on-the-fly updates and visualizations for time-series data and sensor metrics.
- Optimized **Java** backend systems for performance and scalability, using **multithreading** and **Java's CompletableFuture** for improved concurrency, which reduced response times and enhanced throughput for high-traffic endpoints.
- Led the development of full-stack applications using **React** for the front-end and **Node.js** for the backend, enabling faster feature delivery, streamlined development processes, and improved overall team efficiency by adopting a unified **JavaScript** stack.
- Implemented **Alembic** for database schema management in a complex, multi-database environment. This automated schema versioning and rollback procedures, reducing errors during application updates and simplifying deployment processes.

May 2016 – May 2021

Full Stack Engineer, IBM **|** location: Doral, FL

- Led the development of scalable, high-performance applications using **Python**, **Django**, and **Flask**, improving system efficiency and reducing processing time.
- Designed and developed RESTful APIs using **Flask** and **FastAPI**, enabling seamless integration with front-end applications and third-party services.
- Optimized microservices architecture by integrating Gunicorn as the **WSGI HTTP server** for a **Python** web application, fine-tuning worker configurations to handle varying traffic loads and reducing latency during peak usage times.
- Integrated **React** and **Node.js** into an existing Python-based backend architecture, improving user experience with dynamic front-end updates and ensuring smooth, real-time data exchanges via RESTful APIs and GraphQL endpoints.
- Architected and developed **GraphQL APIs** with Python (Graphene), allowing for flexible, efficient data querying and improving frontend performance by enabling real-time, dynamic data fetching based on user interactions.
- Designed and implemented high-performance **PostgreSQL** database architectures, optimizing query performance through effective indexing strategies and ensuring data integrity for critical applications with complex data models.
- Introduced Confluence as a central knowledge-sharing platform, creating and maintaining technical documentation, release notes, and team guides, which improved **cross-team communication** and **accelerated onboarding** for new developers.
- Integrated complex business logic into the **ORM** layer, enabling efficient data validation and seamless interaction between **Python code** and **PostgreSQL**, while leveraging **SQLAlchemy's** declarative mapping for better maintainability.
- Developed cloud-native applications on AWS, leveraging **EC2**, **S3**, **RDS**, and Lambda to build scalable and high-performance systems that enhanced application availability and significantly lowered infrastructure costs.
- Streamlined third-party integrations by automating **OAuth2** client configuration and token issuance, making it easier to manage tokens in dynamic environments and improving the onboarding process for new integrations.
- Implemented robust **MySQL** replication and failover strategies, ensuring high availability and fault tolerance for critical applications, which helped minimize database downtime during both planned and unplanned maintenance events.

- Designed and built a custom Buildkite agent to automate deployment pipelines, optimizing build times through parallelism and better resource allocation, resulting in faster and more efficient deployments.
- Improved codebase maintainability by using **dependency injection** to replace hardcoded service initialization, enabling better configurability and the ability to inject custom implementations at runtime.
- Created end-to-end **CI/CD** pipelines using **Jenkins, GitLab CI, and CircleCI**, automating build, test, and deployment workflows to significantly reduce release times and improve development cycle efficiency.
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- Automated database migrations and schema changes using **Alembic**, ensuring smooth updates to the Postgres schema and consistency across different environments.

Jun 2013 – Feb 2016 **Software Developer,** Collabera | location: Doral, FL

- Developed responsive, cross-browser web interfaces using **HTML5, CSS3**, and **JavaScript**, ensuring seamless user experiences across desktop, tablet, and mobile devices. Applied best practices for web performance, which helped improve user engagement and satisfaction metrics by reducing page load times and increasing interactivity.
- Introduced progressive enhancement strategies for legacy systems, upgrading front-end code by implementing HTML5 semantic elements and utilizing modern CSS3 flexbox and grid layouts. This effort improved accessibility for all users, including those with disabilities, and ensured the site was fully usable across various devices, particularly mobile.
- Engineered secure authentication solutions using **JWT** (JSON Web Tokens) to create stateless, scalable systems for both web and mobile applications. This simplified the authentication flow, reducing dependency on server-side sessions and enabling smoother, more secure user login processes.
- Designed and deployed **Python-based AWS Lambda functions** integrated with **AWS API Gateway**, allowing efficient communication between front-end applications and back-end services. This architecture improved system performance by decreasing latency and increased reliability, helping to scale the application without managing dedicated servers.
- Architected a scalable, high-performance microservices architecture for **a large-scale e-commerce platform** using Python, AWS Lambda, API Gateway, and **SQS**. This system handled millions of daily requests, significantly improving the platform's uptime, responsiveness, and overall customer experience by optimizing resource allocation and load balancing.
- Built and maintained CI/CD pipelines for both JavaScript and Python-based projects, utilizing **Jenkins, Docker**, and **Kubernetes**. These pipelines streamlined the deployment process, enabling more frequent releases with fewer errors and reducing the time needed to push features to production.
- Implemented custom **DataDog** agent extensions in Python to monitor legacy applications within hybrid cloud environments (both on-premises and AWS). These extensions allowed for real-time performance tracking and issue identification, which dramatically reduced the troubleshooting time and improved overall system stability.

EDUCATION

Bachelor's degree in Computer Science
FLORIDA STATE UNIVERSITY | TALLHASSEE, FL 2012

STRENGTHS

- Thrive on challenge, Enjoy challenging, Goal-oriented, Full of idea, Smart work , Self-starter, Self-disciplined, Team work, Adapting new environment, Communication